Class 6

Biology

Olympiad Program
Volume - I

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1. Food - Where Does It Come From?

Solutions

LEVEL-I

1. Correct option (D)

A condiment is a food item that is added to another food item to enhance its flavor. Due to its widespread usage in enhancing food flavor, pepper is regarded as a condiment. Pepper is a flavorful spice that is used in many different cuisines to add heat, depth, and complexity to foods. The Piper nigrum plant's dried, crushed, or powdered peppercorns are used to make it.

Pepper is one of the most well-known and adaptable spices in the world and has been used as a condiment for ages. It gives a food a unique flavor and can be a key element in creating the appropriate flavor profile.

2. Correct option (D)

Ingredients, cooking technique, and regional cultural customs are just a few of the variables that affect how food tastes.

It goes without saying that a dish's flavor will be greatly influenced by the items utilized. For instance, a food cooked with soy sauce, ginger, and sesame oil will taste considerably different than a dish made with tomatoes, basil, and garlic.

The way food is prepared can also impact how it tastes. For instance, a rare steak will taste quite different from a well-done steak.

The regional cultural customs can have an impact on how food tastes. For instance, while adding spices to meals is prevalent in certain cultures, it is less frequent in others.

3. Correct option (D)

When purchasing packed food, it is important to consider all of the following:

a) Ingredients: Checking the list of ingredients is crucial to understand what the food product contains. This helps in determining if there are any specific ingredients you may want to avoid due to allergies, dietary restrictions, or personal preferences.

b) Manufacturing date: The production date tells us when the product was created. In particular for perishable commodities, it might be helpful for determining the freshness and shelf life of the product. The production date may be less important for non-perishable commodities, though.

c) Expire: The recommendation for when the product should be consumed is indicated by the expiration date. It assists in determining the food's safety and freshness. Foodborne infections might result from consuming food after it has expired, which is dangerous.

4. Correct option (B)

Almonds are not typically considered a fragrant material. Almonds do have a light nutty flavor, but unlike the other items on the list, they are not particularly well-known for their robust scent.

A) Cinnamon: This spice is well-known for its distinctive, pleasantly warm, and sweet fragrance. It is frequently used as a spice and gives food, drinks, and baked items a lovely scent.

C) Cloves: The scent of cloves is powerful, fragrant, and spicy. They are frequently employed in baking and cooking to provide a rich, cozy, and slightly sweet scent to food.

D) Cardamom: This spice has a distinct aroma that is both sweet and somewhat spicy. Cardamom is quite aromatic. With its pleasantly fragrant flavor, it is frequently used in both savory and sweet recipes.

5. Correct option (D)

All of the techniques described - sugar syrups, honey, and refrigeration - may be utilized for fruit preservation. There are several ways to preserve fruits.

Sugar syrups: By submerging fruits in sugar syrups, which helps maintain their flavor, texture, and color, fruits can be preserved. Inhibiting the growth of germs that might contaminate the fruit is how the sugar syrup functions as a preservative. Fruits including cherries, peaches, pears, and pineapples are frequently preserved using this technique.

Honey: Honey may also be used as a natural fruit preserver. By fostering an environment that is unfavorable for microbial development, it aids in preventing spoiling. Honey may be used to preserve fruits by coating them in it or by preparing jams and preserves with it.

Refrigeration: Fruits may be kept in their fresh form for a longer amount of time by using refrigeration. The fruit's shelf life is increased by lowering the temperature since it inhibits microbial development and enzymatic activity. Fruits may be kept fresh and maintained for a number of days or weeks by being stored in the refrigerator, depending on the fruit variety.

6. Correct option (D)

Root: The root of several plants, including carrots, radishes, beets, and potatoes, can be the only edible portion. Because of their nutritional value, these roots are frequently used in many culinary dishes as vegetables.

Stem: Some plants have edible stems that are commonly consumed as food. Examples include asparagus, celery, broccoli, and bamboo shoots. These stems are often cooked or used raw in salads and stir-fries.

Leaves: Many plants contain edible leaves that may be utilized as either vegetables or herbs. The following are some examples: lettuce, spinach, kale, cabbage, basil, and mint. Both raw and cooked meals may be made with these leaves.

Therefore, depending on the particular plant species and the culinary customs of various cultures, the roots, stems, leaves, fruits, and seeds of plants can all be used as food.

7. Correct option (A)

8. Correct option (B)

The staple cuisine of Rajasthan, a desert state, is roti, a sort of flatbread prepared from wheat flour. It is a common dish in Rajasthan and is eaten with a number of other cuisines, including dal, sabzi, and curry.

The flatbread known as chapati is also created from whole wheat flour. Although it is a widespread dish in many regions of India, it is less frequent in Rajasthan than roti.

In Rajasthan, rice is not a common meal. The southern states of India, where the environment is more humid, are where it is more prevalent.

Sambar and idli are South Indian foods. In Rajasthan, they are not frequently consumed.

9. Correct option (D)

Major food crops grown in India include jowar, wheat, and rice.

In India, rice is the most significant crop for food production. Although it is cultivated across the nation, the eastern and southern areas are where it is most significant.

In India, wheat is the second most significant food crop. It is cultivated in the northern and western states since they have more favorable climates for growing wheat.

A kind of millet called jowar is cultivated in the central and western regions. It is crucial in regions with unpredictable rainfall since it is a drought-resistant crop.

Approximately 70% of India's entire food grain production is comprised of these three crops.

10. Correct option (B)

Indian food frequently uses the strong spice asafoetida to add flavour. It belongs to the same family as celery and carrots, the Umbelliferae. It is made from the resinous sap of the roots of Ferula species of plants, mainly Ferula assa-foetida. The plant's thick taproot is used to extract the sap, which is then dried to make resin that is either powdered or utilised in its raw state. Asafoetida is not commonly made from the plant's stem, leaves, or blossoms.

Indian food uses asafoetida, particularly in pickles, curries, and lentil meals. Traditional medicine also uses it to treat colic, indigestion, and gas.

11. Correct option (A & B)

All domestic chickens are members of the genus Gallus, which also contains the red junglefowl. Male chickens are called roosters, while female chickens are called hens. Food, eggs, and feathers may all be obtained from hens and roosters.

Goats are not chickens. They are animals belonging to the Capra genus, which also contains antelopes, sheep, and goats. Normally, goats are not utilized for food, eggs, or feathers.

12. Correct option (D)

We get our minerals from a variety of sources, including plants, animals, and the sea.

Numerous minerals, including calcium, iron, magnesium, potassium, and zinc, are abundant in plants. Leafy green vegetables, legumes, nuts, seeds, and whole grains are a few of the greatest plant-based sources of nutrients.

Animals are a good source of minerals including iron, zinc, selenium, and iodine. Meat, poultry, fish, eggs, and dairy products are some of the richest sources of minerals from animals.

Minerals including iodine, selenium, and magnesium is abundant in seawater. Seafood, seaweed, and salt are a few of the richest sources of minerals found in the ocean.

13. Correct option (A & B)

Both Spinach and coriander plants feature edible leaves that may be used in a number of ways.

Leafy greens like spinach are a good source of vitamins A, C, and K. It is an excellent source of fiber and iron as well. Eaten raw in salads or cooked in a variety of ways, spinach leaves are versatile.

The herb coriander is a common ingredient in international cuisines. Coriander leaves are used to flavor curries, chutneys, and other foods because of their pungent, lemony flavor. Tea and food garnishes may both be made with coriander leaves.

Acacia leaves are rarely used in cooking and are not edible.

14. Correct option (A & B)

Two of the most popular dishes in South India are rice and dal. The primary diet in South India is rice, which is eaten with a variety of foods including dal, sambar, and chutneys. A well-liked food in South India is dal, a lentil soup.

The flatbread known as Phulka is not as popular in South India as rice and dal. Phulka is frequently eaten with curries and other foods and is primarily prepared from wheat flour.

Rice and dal are the two food categories that are most popular in South India. Therefore, D is the correct response.

15. Correct option (D)

16. Correct option (a-Q, b-T, c-R, d-P, e-S)

a) Boiling - Q) Potatoes, Eggs

Food products are cooked by immersion in hot water or another liquid during the boiling process. To get the proper amount of doneness, potatoes and eggs are frequently cooked. As a result, Group A includes potatoes and eggs.

b) Roasting - T) Chicken, fish

Cooking food in an oven or over an open flame while typically employing dry heat is known as roasting. For better flavor and a crispier texture, fish and chicken are frequently roasted. Consequently, Group B includes fish and chicken.

c) Chopping - R) salads

Food products are sliced into smaller, more manageable pieces by the process of "chopping." Salads can contain a wide range of items, but typically, vegetables and other salad elements are prepared by being chopped. As a result, Group B includes salads.

d) Fermentation - P) Bread, Cakes

Sugars are transformed into alcohol or organic acids during the fermentation process by the activity of yeast, bacteria, or other microorganisms. Food products that go through fermentation include bread and cakes, which often use yeast as a leavened. Therefore, Group B includes bread and cakes.

e) Steaming - S) idle

When food is cooked by steam produced by boiling water, it does not come into direct contact with the liquid. Indian traditional food known as idle is prepared by steaming a fermented batter largely comprised of rice and black lentils. Idle is therefore in Group A.

LEVEL-II

1. Correct option (A)

Nutrients, which are helpful components found in food, are crucial for the healthy operation and upkeep of the body. Water, carbs, proteins, lipids, vitamins, and minerals are all considered nutrients. These compounds stimulate development and repair, supply energy, control body functions, and promote general health and wellbeing.

The capacity to perform work is energy. When we eat food, our bodies break down the nutrients into smaller molecules that can be used for energy.

Weight is a unit of measurement for an object's weight. Nutrients don't include weight. It is only a measurement of an object's mass.

Energy-giving nutrients include carbohydrates, which are a type of nutrient. Bread, pasta, grains, fruits, and vegetables, among other foods, contain them.

2. Correct option (B)

The components of food known as nutrients are what our bodies require to remain healthy. They provide us energy, support our physical and mental development, and keep us healthy. Our immune system, which is our body's line of defense against sickness, may be strengthened by certain foods including vitamins and minerals.

A strong immune system is necessary for illness prevention. Our immune system can fend against pathogens and viruses that can make us ill while it is healthy. Vitamin A, C, D, zinc, and selenium are nutrients that support a healthy immune system.

Other nutrients, such as the antioxidants in fruits and vegetables, can aid in lowering the risk of chronic illnesses including diabetes, cancer, and heart disease. As a result, eating a balanced, nutrient-rich diet is crucial for preventing disease and sustaining good health.

3. Correct option (A)

Paddy is the unhusked grain of rice. It is the most common type of rice used in cooking. To cook rice, we need to wash the paddy and then cook it in water. The amount of water we need to use depends on the type of rice we are using. The husk is subsequently removed from these grains during processing, leaving just uncooked rice. When preparing rice, we mix raw rice with water, heat the mixture, and then the rice absorbs the water and cooks.

Wheat, jowar, and rava are not used to cook rice. Wheat is used to make bread, jowar is used to make jowar roti, and rava is used to make rava idle.

4. Correct option (C)

The different items or components used to produce a meal are known as ingredients. They are the precise ingredients or mixtures of ingredients that are mixed together and utilized in a particular recipe or dish in a certain amount or ratio. Among the food elements that affect a dish's flavor, texture, and general make-up are vegetables, fruits, meats, grains, spices, herbs, oils, and other ingredients. They are necessary for cooking and have a significant impact on how good the food turns out.

Items that are not utilized to produce a meal are referred to as non-ingredients. They might be anything from kitchenware and cutlery to cleaning products.

Any object or thing can be referred to by the broad word "items." Items can refer to ingredients, things other than ingredients, or even the meal itself when discussing food.

A prepared meal is specifically referred to as a dish. An ingredient can be used alone or in combination to create a meal.

5. Correct option (A)

The animal's flesh that is consumed as food is known as meat. It contains plenty of iron, protein, and other minerals. Cattle, pigs, lambs, chickens, and fish are just a few of the animals whose meat is available. In many diets, meat is a substantial source of protein, vitamins, minerals, and other nutrients. It is a staple cuisine in many cultures all over the world and is frequently consumed by people as part of their normal diet.

Meat cannot be obtained from plants. Other foods including fruits, vegetables, and grains can be found in them as well.

6. Correct option (D)

Plants are the source of fruits and vegetables. Vegetables include a variety of edible plant components such leaves, stems, roots, and bulbs, whereas fruits are the mature ovaries of blooming plants and carry seeds. They are abundant sources of fiber, vitamins, minerals, and other healthy substances. Humans frequently consume fruits and vegetables because of their nutritional worth and health advantages. They are also an essential component of a balanced diet.

Options A) Oil and Ghee, and B) Milk and Pulses, and C) Ghee and Milk are incorrect because they include animal-derived products (oil and ghee from animals, milk from animals, and pulses being a plant-based source of protein).

7. Correct option (A)

Both the meat and the ghee come from animals. Ghee, a sort of clarified butter that is commonly used in cooking, is produced from butter that has been boiled and the milk particles taken out. Meat is any animal's edible flesh, including that of fish, fowl, pigs, and cattle. These foods, which come from animal origins, are used as food in many cultures and provide nutrients including protein, fats, vitamins, and minerals.

Option B) Fruits and Vegetables and Option C) Dry fruits and Ghee are incorrect because fruits, vegetables, and dry fruits are obtained from plants, not animals.

8. Correct option (D)

Cereals are the grains of certain grasses that are used for food. They are a good source of carbohydrates, fiber, and vitamins. Some common cereals include rice, wheat, maize, millets, barley, and oats.

The two grains that are most extensively grown worldwide are rice and wheat. For many people all across the world, they are a staple meal. Another important cereal crop is maize, which is used to produce a range of goods including cornmeal, corn flour and corn oil. Small, hard-grained grains known as millets are cultivated in warm areas. They are frequently used to produce porridge and bread, and they are a rich source of protein and fiber. Beer, bread, and breakfast cereals all use the cereal barley. Making muesli and muesli using oats, a grain rich in protein and fiber, is quite common.

9. Correct option (A & B)

Both coconut and mustard seeds may be used to make oil.

A) Mustard, Coconut - Mustard oil, which is extensively used in cooking and other culinary purposes, is typically extracted from mustard seeds. Coconut oil, which is frequently used in baking, cooking, and cosmetics, is extracted from coconut seeds.

B) Coconuts, sesame - As previously indicated, coconuts are utilized to obtain coconut oil. Sesame oil, sometimes referred to as gingelly oil or sesame oil, is extracted from sesame seeds and used in cooking, salad dressing, and flavoring.
C) Guava, Sunflower, on the other hand, is inaccurate because guavas are fruits and aren't normally utilized to produce oil. However, sunflower oil is frequently extracted from sunflower seeds.

10. Correct option (D)

Spices are indeed used for various purposes, including flavoring, coloring, and preservation.
a) Flavoring: Spices are widely used to enhance the taste, aroma, and overall flavor profile of dishes. They add depth, complexity, and distinct flavors to culinary preparations.
b) Coloring: Some spices, such as turmeric, paprika, saffron, and annatto, are used to provide natural coloring to food. These spices can impart vibrant hues and enhance the visual appeal of dishes.
c) Preservation: Certain spices possess antimicrobial properties that can help inhibit the growth of harmful bacteria or microorganisms. As a result, they are used to some extent in food preservation to extend shelf life or prevent spoilage.

11. Correct option (A)

The seeds of the Cuminum cyminum plant are dried and used to make the spice cumin. It is a Mediterranean and Middle Eastern native and a member of the parsley family. Indian, Middle Eastern, and Mexican cuisine frequently employ cumin because of its powerful, earthy flavor. Additionally, it's a component of certain stews and sausages from Europe.
Spices are not sugarcane or jaggery. A plant called sugarcane is used to produce sugar. Sugar that is manufactured from sugarcane is called jaggery. In contrast to spices, they are not used to flavor food.

12. Correct option (C)

The leaves of the tea plant, Camellia sinensis, are used to make tea. To make several kinds of tea, including black tea, green tea, oolong tea, and white tea, the young leaves and leaf buds of the tea plant are plucked and processed. To create their distinctive flavors, the leaves are dried and put through a variety of oxidation and fermentation procedures.
On the other hand, coffee is made from coffee beans, which are the Coffea plant's seeds. The fruit, which is frequently referred to as coffee cherries, contains the coffee beans. To bring out the distinctive flavors and fragrances of coffee, coffee cherries are selected, and then the seeds are removed and roasted.
Therefore, the correct answer is C) Leaves and Seeds - Tea is obtained from the leaves of the tea plant, and coffee is prepared from coffee seeds (beans).

13. Correct option (B)

Plants prepare their own food through the process of photosynthesis. Photosynthesis is a biochemical process that occurs in the chloroplasts of plant cells, primarily in the leaves. It involves the conversion of light energy from the sun into chemical energy in the form of glucose (sugar). During photosynthesis, plants use carbon dioxide from the atmosphere, water from the roots, and sunlight to produce glucose and oxygen.
When an organism breathes, oxygen is used to break down glucose and release energy. All live cells go through this procedure.
The process through which plants lose water via their leaves is called transpiration. This procedure circulates water and nutrients throughout the plant while also aiding in cooling it.
The process through which blood is circulated around the body is called circulation. This procedure is necessary for delivering nutrients and oxygen to the cells as well as for eliminating waste.

14. Correct option (C)

Regarding food chains and ecosystems, plants are viewed as producers. Through the process of photosynthesis, as was previously established, they have the capacity to produce their own food. Producers are living things that can transform light energy into chemical energy by using sunlight and inorganic elements (such carbon dioxide and water) to create organic molecules, mainly glucose. Other creatures in the food chain can obtain energy from these chemical molecules.

Consumers are living things that can't cook for themselves. To obtain the necessary energy, they must consume other living things. Herbivores, carnivores, and omnivores are all types of consumers. Creatures that decompose dead creatures are called decomposers. In order for producers to utilize the nutrients from the deceased organisms, they release them back into the environment.

Therefore, plants are only producers.

15. Correct option (A)

Plants store food primarily in the form of starch. Starch is a complex carbohydrate made up of glucose units that are linked together. It serves as the main energy storage molecule in plants. During photosynthesis, excess glucose produced is converted into starch and stored in various plant parts, such as roots, tubers, seeds, and fruits. Starch is insoluble in water, which makes it an efficient and compact storage form for plants to store energy for future use.

Although plants do contain proteins, they are not the principal food storage type in plants, hence Option B) is wrong.

Option C) is false because, although being a mineral necessary for plant growth and development, calcium is not the major way that plants store food.

16. Correct option (D)

Carrots are a type of root vegetable. Plants that produce root vegetables store nourishment in their roots. The carrot root is a taproot, which means it descends directly into the ground. The carrot plant's primary storage organ is the taproot.

The root system of the carrot plant stores nutrients and energy, enabling it to develop and survive under unfavourable conditions.

Because of its sweet and crisp flavour and nutritional content, the carrot plant's root is th e component that is frequently eaten as a vegetable. It holds vitamins, proteins, and carbs.

Carrot leaves have a green, fluffy appearance. The carrot root's crown is where they are connected. The leaves assist the carrot plant's photosynthetic process and food production.

17. Correct option (A)

Apples are a type of plant's fruit. The mature ovary of the apple tree's blossom is what we know as an apple. It grows from the fertilized blossom and contains apple tree seeds. Apples come in a variety of colors, shapes, and flavors. The apple fruit is a good source of vitamins and minerals. It is also a good source of fiber. Apples are a popular fruit that can be eaten fresh, cooked, or juiced.

Option B) Leaf is wrong since it is not the portion of the plant that produces the apple fruit; rather, it is a distinct plant component engaged in photosynthesis and gas exchange.

Option C) Because apples are officially categorized as fruits and not leaves, Options A and B are equally erroneous.

18. Correct option (A)

The edible portion of the black gramme plant is the seed, which is a sort of pulse or legume. Small, oval-shaped legumes known as black gramme seeds, urad dal, or black lentils are frequently used in a variety of dishes around the world. Black gramme seeds are an

important component of vegetarian and vegan diets because they are high in protein, fiber, and other minerals.

Black gram is not a fruit, hence Option B) Fruit is wrong.

Both Options C) Leaf and D) Root are inaccurate since they do not depict the section of the black gram plant that is edible.

19. Correct option (A)

Milk is primarily obtained from cows and is commonly consumed as a beverage. Cows are widely domesticated for milk production, and their milk is a rich source of essential nutrients such as protein, calcium, vitamins, and minerals. Milk is consumed in its natural form or used as an ingredient in various dairy products such as cheese, yogurt, butter, and cream. While there are other animals that produce milk, such as goats, sheep, and buffaloes, the most commonly consumed and widely available milk is derived from cows.

Option A) Cat and Option C) Cat are inaccurate because, although cats may produce milk to feed their kittens, humans do not normally ingest it on a daily basis.

20. Correct option (B)

Rearing and management of fish on a large scale is known as pisciculture. Pisciculture is the practice of breeding, rearing, and cultivating fish in ponds, tanks, or other controlled aquatic environments. It involves activities such as stocking fish, providing suitable conditions for their growth and reproduction, managing water quality, feeding, and harvesting. Pisciculture is an important practice for commercial fish production, aquaculture, and meeting the demand for fish as a food source.

The other options are incorrect. Poultry is the raising of domesticated birds for food. Domestication is the process of taming and breeding animals for human use. Agriculture is the practice of cultivating plants and livestock.

21. Correct option (D)

Poultry is the general term for domesticated birds such hens, geese, quails, and ducks. In poultry farming, the birds are raised and managed for a variety of uses, including the production of eggs and meat (poultry meat). The agriculture sector's key sector is poultry farming, which is crucial in supplying the world's need for chicken products. Poultry is a major source of protein for humans. It is also a good source of other nutrients, such as vitamins, minerals, and iron.

The other options are incorrect. Pisciculture is the practice of rearing and managing fish on a large scale. Agriculture is the practice of cultivating plants and livestock. Dairy Farm is a place where dairy cows are raised for milk.

22. Correct option (C)

A cow is an example of an herbivorous animal. Animals known as herbivores are those that eat predominantly plant-based diets including grass, leaves, and other types of flora. Cows' specialized digestive systems, which include many chambers in the stomach, enable them to effectively digest and extract nutrients from plant matter. They have developed to rely on an herbivorous diet and can feed on grass and other plant materials with ease.

Carnivores are animals that eat meat. They have sharp teeth and claws that help them to catch and kill their prey. Examples of carnivores include lions, tigers, and wolves.

Omnivores are animals that eat both plants and meat. They have a variety of teeth and digestive systems that allow them to eat both types of food. Examples of omnivores include humans, bears, and pigs.

23. Correct option (B)

Evaporation is the method used to produce salt from seawater. Various dissolved salts may be found in sea water, with sodium chloride (NaCl) being the most prevalent. Sea water is gathered and allowed to evaporate to produce salt, either naturally through solar

evaporation or artificially. The leftover salt crystals are gathered when the water evaporates, concentrating the salt content. Salt is a valuable commodity that has been used by humans for centuries. It is used in cooking, preserving food, and making soap and other products.
Option A) pure water is wrong since it is uncommon to extract salt from sources of pure water. Compared to sea water, fresh water sources have lower salt content.
Option C) Polluted water is inaccurate as well since it is not a reliable or appropriate supply of salt.

24. Correct option (B)

Omnivores are organisms that have the ability to consume and derive nutrients from both plant-based and animal-based sources
Human and Cow - Humans are omnivorous, eating both plant and animal foods in their diets, but cows are herbivorous, eating mostly plant-based meals. This choice is thus unsuitable.
The Human and the Crow – Both people and crows are omnivorous. While crows are opportunistic feeders who eat a range of items, including fruits, seeds, insects, and small animals, human diets are broad and contain both plant and animal foods. So, this choice is the right one.

25. Correct option (A)

Animals that feed on the flesh of dead animals are called scavengers. Scavengers play an important ecological role by consuming and disposing of decaying organic matter. They help to clean up the environment by recycling nutrients from dead animals back into the ecosystem. Examples of scavengers include vultures, hyenas, crows, and certain species of beetles.
Option B) Parasites is incorrect because parasites are organisms that live in or on another organism, called the host, and obtain nutrients from the host's body. Parasites may feed on the host's tissues or body fluids, but they are not specifically associated with feeding on the flesh of dead animals.
Therefore, the correct answer is C) Only A - The animals that feed on the flesh of dead animals are called scavengers.

26. Correct option (B)

An insect parasite is a mosquito. In particular, female mosquitoes are hematophagous, meaning they consume other creatures' blood, including human blood. They penetrate the skin with their unique mouthparts to draw blood, which provides nutrients for the development of the egg. Female mosquitoes need a blood meal in order to lay eggs; male mosquitoes, on the other hand, mostly consume nectar and plant liquids.
Option A) Scavengers is inaccurate since mosquitoes do not feed on decaying organic debris or dead animals, whereas scavengers do.
Option C) Decomposers is inaccurate since decomposers are creatures that help in the decomposition process by breaking down dead organic matter into simpler forms. Mosquitoes don't contribute much to decomposition.
Because they feed on the blood of other creatures, mosquitoes are regarded as parasites, hence the correct response is B).

27. Correct option (A)

Endoparasites are living things that reside inside another creature. They can lead to a number of disorders and are frequently tiny. Endoparasites are parasites that reside within the body of their host. They obtain their nutrition and other resources by feeding on the tissues, fluids, or blood of the host. Tapeworms, roundworms, and parasites that cause malaria are a few examples of endoparasites.

Ectoparasites, in contrast, exist outside of another organism's body. They may also spread illness and are frequently bigger than endoparasites. Ectoparasites include things like fleas, ticks, and lice.

Organisms that decompose dead organic substances are called decomposers. They return nutrients to the environment, where other creatures can utilize them. Among the decomposers are fungus and bacteria.

Organisms that can produce their own nourishment are known as producers. They produce molecules that are high in energy, like glucose, using sunlight, water, and carbon dioxide. Examples of producers include some bacteria, algae, and plants.

28. Correct option (A)

Peacocks are indeed omnivorous birds, meaning they consume both plants and animals. While their diet consists mainly of plant matter such as seeds, fruits, and grains, they also opportunistically feed on insects, small reptiles, and other small animals. Peacocks are known to display a varied diet, making them omnivores.

The first option, "herbivorous," is untrue because peacocks eat a variety of foods. They are omnivores because they eat both insects and small animals in addition to plant-based meals like seeds and fruits, making them omnivores.

Option B) Carnivorous: This choice is untrue since peacocks do not eat meat or flesh as their main source of food. Even though they occasionally eat small animals, they do not eat primarily carnivorous foods.

Producer (Option D): This choice is untrue since producers are living things with the capacity to grow their own food through processes like photosynthesis. Being animals, peacocks must rely on outside food sources because they are unable to manufacture their own food.

29. Correct option (B)

Through mechanisms like photosynthesis, animals are unable to create their own nourishment. They rely on eating other living things or organic material to get the food and energy they require to survive. As a result, in terms of ecology, animals are considered consumers. species that consume other species or their products are known as consumers. They get their energy from the food they eat, which they also digest.

Producing your own food through processes like photosynthesis or chemosynthesis requires creatures like plants, algae, or certain bacteria, which are not producers. Therefore, Option A) Producer is false.

Option C) Decomposer is inaccurate since decomposers are creatures that help in the decomposition process by breaking down dead organic matter into simpler components.

Option D) None is inaccurate because it is true that animals are considered to be consumers.

30. Correct option (B)

From poultry, which refers to domesticated birds like chickens, we obtain eggs and meat.

Eggs are a typical and popular dietary item made from fowl. Eggs are harvested from chickens that have laid eggs for human use. Eggs are adaptable and utilized in a variety of culinary dishes.

Another thing you may get from fowl is meat. The flesh of chickens and other poultry animals is grown for human consumption. In many cuisines, poultry meat—such as chicken meat—is a substantial source of protein.

Option A is untrue since milk is not produced by using chickens. Most animals, including cows, goats, and sheep, provide milk.

Option C) is untrue since honey is not produced by poultry, which makes egg and honey inaccurate. Honey is created by honey bees from floral nectar.

Option D) Egg and fish is incorrect because fish is not obtained from poultry. Fish is a separate category of aquatic animals, not related to poultry.

2. Components of Food
Solutions

LEVEL-I

1. Correct option (C)

The numerous nutrients included in food ingredients are substances that our bodies require for normal operation and good health. Carbohydrates, proteins, lipids, vitamins, minerals, and water are all examples of nutrients. Each nutrient performs a specialized function in the body, such as supplying energy, boosting the immune system, repairing and developing tissues, and preserving numerous physiological functions.

Carbohydrates, proteins, and fats are all nutrients. However, they are not the only nutrients that the body needs. Other important nutrients include vitamins, minerals, and fiber.

2. Correct option (D)

The common forms of carbohydrates found in food are starch and sugars.

A) Starch: Starch is a type of complex carbohydrate that is present in foods such as grains (such as rice and wheat), tubers (such as potatoes), and legumes (such as beans and lentils). It is a kind of energy storage in plants made up of extended chains of glucose molecules joined together.

B) Sugars: Sugars are simple carbohydrates that come in a variety of forms, including fructose from fruits, sucrose from table sugar, lactose from milk, and honey. They are made up of one or two sugar molecules and offer the body a rapid source of energy.

C) Glucose: A kind of sugar, glucose is not the most prevalent form of carbohydrates present in meals, despite being the body's main source of energy.

3. Correct option (B)

Proteins are regarded as nutrients that help build muscle since they are essential for the development, maintenance, and repair of bodily tissues. Amino acids, the body's building components, make up what they are. The growth and repair of muscles, organs, skin, hair, nails, and other bodily tissues depend on proteins.

Although they may create energy, proteins do not have this as their main purpose. The primary elements that provide the body energy are carbs and fats. Proteins are crucial for general growth and health because they play a major role in structural and functional activities. So, "Body-building foods," option B, is the appropriate response.

4. Correct option (A)

The Biuret test, which is used to check for the presence of proteins, is the procedure you described. A food item that includes proteins becomes violet when copper sulphate and sodium hydroxide are added to it. This is as a result of the proteins' reaction with copper sulphate to create a complex that has a violet color.

When added to the test solution, fats and starches will not cause the test solution to become violet because they do not react with copper sulphate and sodium hydroxide.

5. Correct option (C)

A balanced diet is one that has all the necessary nutrients in the appropriate amounts. This implies that it has the proper amounts of carbs, lipids, proteins, vitamins, minerals, and roughage.

The body needs all of the key nutrients—carbohydrates, lipids, proteins, vitamins, and minerals—for effective operation and general health.

Proteins, lipids, and carbohydrates are macronutrients that support diverse physiological functions and provide us energy.

Micronutrients like vitamins and minerals are necessary in lesser amounts but are just as crucial for many body processes. These processes include supporting metabolism, preserving a robust immune system, and ensuring that enzymes and hormones work as intended.
The two categories of micronutrients that are listed in Option C, "Vitamins & Minerals," are accurately identified.

6. Correct option (C)

A balanced diet is one that has all the necessary nutrients in the appropriate amounts. This implies that it has the proper amounts of carbs, lipids, proteins, vitamins, minerals, and roughage.
For the purpose of sustaining excellent health, a balanced diet is crucial. Numerous health issues, including as obesity, heart disease, stroke, and cancer, can be avoided with its aid. Energy levels, happiness, and general wellbeing can all be helped by it.
When the body has too much fat, the condition is called obesity. It results from consuming too many calories and not working out enough.
A diet that lacks all the necessary nutrients in the proper amounts is considered deficient. This may result in a number of health issues, such as weariness, weakness,

7. Correct option (A)

A form of fiber known as roughage is not digested by the body. It can help lower the risk of constipation and maintain the health of the digestive system. By giving the stool more weight, roughage aids in the removal of undigested food from the body by causing the intestinal walls to contract. Food is pushed through the digestive system by this motion.
The body need vitamins, minerals, and proteins among other vital components to function effectively. They are not, however, broken down by the body the same way as roughage is. The small intestine is where vitamins, minerals, and proteins are absorbed into the circulation. Roughage does not enter the circulation; rather, it travels through the digestive system and is expelled from the body.

8. Correct option (B)

Thiamine (vitamin B1) insufficiency leads to the illness beriberi. This is because thiamine is essential for the proper function of the nervous system and muscles. When the body does not get enough thiamine, the muscles can become weak and fatigued. It mostly affects the neural system and can cause a variety of symptoms, such as muscular weakness and atrophy, trouble walking, and problems with the nerves. The word "beriberi" is derived from the Sinhalese word for "weakness."

9. Correct option (C)

Although minerals are necessary nutrients for the body, they are only required in modest amounts when compared to macronutrients like carbs, proteins, and fats. Minerals are essential for a number of physiological activities, including bone health, neuron function, enzyme activity, and maintaining a healthy fluid balance, despite the fact that their requirements are quite low.
The number of minerals required for the body's optimal operation is best summed up by Option C, "Small amounts." Like macronutrients, minerals are not required in huge quantities, but their presence is essential to maintain general health and wellbeing.

10. Correct option (A)

Carbohydrates and fats are both macronutrients that serve as a significant source of energy for the body. They are crucial for preserving a healthy weight since they are the body's primary source of energy. Glucose, the body's primary energy source, is created when carbohydrates are broken down. Fatty acids are created when fats are broken down, and they may either be utilized right away for energy or stored for future use.

Carbohydrates are more easily and quickly converted into energy, while fats provide a more sustained and longer-lasting source of energy.
Protein-rich diets are foods that are good for growing muscle. Building and repairing bodily tissues depend on protein. Foods that demand energy for digestion are called energy consumers. High-protein foods and foods high in fiber are among these. Foods that are protective contain vitamins, minerals, and other elements that guard the body from illness. These include whole grains, fruits, and vegetables.

11. Correct option (D)

Diseases associated with vitamin deficiencies are brought on by a sustained shortage of certain nutrients in the diet. Numerous deficiency disorders can develop when the body doesn't get enough important nutrients for an extended length of time, such as vitamins, minerals, proteins, or lipids. Each vitamin is essential to the body's operation, and shortages or particular health problems can occur from its absence or insufficient ingestion. For instance, a shortage of vitamin C can result in scurvy, a lack of iron can result in anemia, and a lack of calcium can cause brittle bones and osteoporosis. To avoid deficiency disorders, it is crucial to maintain a balanced diet with a range of foods.

12. Correct option (D)

A food item that includes starch turns blue-black when iodine solution is added. This is due to the blue-black complex that is created when starch and iodine react.
Protein and glucose do not react with iodine solution, therefore adding them does not cause them to become blue-black. So, you may add a few drops of iodine solution to a food item to check for the presence of starch. The food item includes starch if the color turns to blue-black.
Glucose and protein do not react with iodine solution, so they will not turn blue-black when added to iodine solution.

13. Correct option (C)

Minerals include calcium, phosphorus, and iron. The body requires trace amounts of minerals, which are inorganic elements, to carry out a variety of physiological processes. They are crucial for sustaining general health since they are involved in various activities like iron transfer, enzyme activity, bone health (calcium and phosphorus), and oxygen transport. Although the body also needs vitamins, carbs, and roughage (fiber), the ideal response in this situation is C) Minerals.

14. Correct option (A)

Numerous nutrients, such as iron, zinc, magnesium, and selenium, are abundant in meat. These minerals are necessary for a number of body processes, including the creation of blood cells, energy synthesis, and immunological response.
Although they may not have as much diversity of minerals as meat, rice, wheat, and bananas are nonetheless excellent providers of several minerals.

15. Correct option (B)

The body requires a lot of nutrients called macronutrients. They consist of proteins, lipids, and carbs. Proteins and fats are necessary for the body's normal operation. They give out energy, support tissue growth and repair, and safeguard organs. Carbohydrates, proteins, fats and water are macronutrients. Vitamins, minerals, roughage are micronutrients.
Here are some of the functions of fats and proteins:
Fats:

- Provide energy
- Help to build and repair tissues
- Protect organs
- Keep the body warm

- Help to absorb vitamins

Proteins:

- Provide energy
- Help to build and repair tissues
- Make enzymes, hormones, and antibodies
- Transport nutrients throughout the body
- Help to regulate blood sugar levels

So, in the given options B) Fats and proteins are the right option in which both are macronutrients.

16. Correct option (C)

Iodine deficiency leads to goiter. The thyroid gland, a butterfly-shaped gland in the neck, enlarges in a condition called goiter. Iodine is a crucial element required for the synthesis of thyroid hormones. The thyroid gland enlarges in an effort to generate more hormones when the body is deficient in iodine, which results in the development of goiter. Iodine supplementation and iodized salt programmers can avoid this condition, which previously was common in areas with low iodine levels in the soil and water.

A is incorrect because iron deficiency does not cause goiter. B is incorrect because silver deficiency does not cause goiter. D is incorrect because salt does not contain iodine.

17. Correct option (B & C)

Due to its crucial function in preserving the health and integrity of these tissues, calcium shortage can lead to weak bones and tooth decay.

i) Tooth decay

Calcium is a necessary mineral that is required for the upkeep of sturdy and healthy teeth. It contributes to the development of the tooth's hard structure, or enamel, which guards the inner layers from acids and bacteria that cause decay. When there is a calcium deficit, the enamel weakens and is more prone to erosion, which causes tooth decay.

ii) Weak bones:

Another important element of bone formation is calcium. 99% of the calcium in the body is kept in the bones and teeth. The body may use calcium from the bones to maintain normal blood calcium levels when there is insufficient calcium intake. This can weaken the bones over time and result in diseases like osteoporosis, increasing the risk of fractures and other accidents.

18. Correct option (D)

Vitamins are referred to as "protective foods" since they are crucial for sustaining general health and shielding the body from illnesses and deficits. They are organic substances that the body needs in little amounts for a variety of physiological processes. Vitamins function as coenzymes, helping enzymes complete crucial chemical processes.

Option A, "Foods that produce energy," and Option C, "Foods that consume energy," are inaccurate definitions of vitamins. Foods that provide energy often relate to macronutrients (carbohydrates, proteins, and fats) that supply calories for energy generation.

Option B (Body-building foods) might refer to proteins, which are macronutrients responsible for building and repairing tissues in the body. While vitamins are essential for various bodily functions, they are not primarily known as body-building foods.

19. Correct option (A)

Vitamins and minerals known as micronutrients are critical for human health. Even though they are just somewhat necessary, they are essential for numerous body processes.

Organic substances known as vitamins are crucial for development, growth, and overall health. Vitamin C and B are examples of water-soluble vitamins, and the other category includes fat-soluble vitamins. vitamin A, D, E, and K

Minerals are inorganic substances that are vital to maintaining good health. They are essential for several biological processes, such as neuron, bone, and blood clotting health. Minerals including calcium, iron, magnesium, phosphorus, potassium, and zinc are among the most crucial ones.

20. Correct option (B)

Scurvy is a condition brought on by a deficiency of vitamin C. Collagen, a protein that provides skin, bones, and blood vessels their strength, is formed only when vitamin C is present. Additionally, it aids in immune system stimulation and infection prevention.

The gums swell and bleed readily when the body is deficient in vitamin C. Additionally, wound healing takes longer. Scurvy also causes joint discomfort, muscular weakness, and weariness.

The illness rickets is brought on by a deficiency of vitamin D. When the body doesn't produce enough healthy red blood cells, it has anemia. Any of these disorders do not manifest as visual loss.

21. Correct option (D)

Our bodies require a balanced diet in order to be strong and healthy. Consuming a range of meals that contain all the necessary nutrients, such as carbs, proteins, fats, vitamins, minerals, and water, is the definition of a balanced diet. Each vitamin supports body processes, cellular development, and general health in a different way.

While certain items like rice (option A) and meat (option B) can be included in a balanced diet, relying only on one or a few particular meals could not provide the body with all the essential elements. Our bodies get the nutrients they need for good health when we eat a variety of meals from various dietary categories. Additionally, it's crucial for a variety of body processes and general health to be appropriately hydrated with water (option C).

22. Correct option (C)

Nutrients are the components that make up various food products. The body need nutrients in order to function effectively. They are broken down into six primary categories: water, carbs, proteins, fats, vitamins, and minerals.

The body's primary fuel source is carbohydrate. They may be found in meals including fruits, vegetables, bread, pasta, rice, and pasta.

The development, maintenance, and repair of the body's tissues depend on proteins. Foods including meat, chicken, fish, eggs, dairy products, and legumes contain them.

For the body to store energy, absorb nutrients, and safeguard organs, fats are necessary. They can be found in foods including nuts, seeds, dairy, meat, chicken, and fish.

Numerous body processes, including eyesight, development, and reproduction, depend on vitamins. Fruits, vegetables, and some animal products all contain them.

Numerous body processes, such as bone health, blood coagulation, and neuron function, depend on minerals. Fruits, vegetables, and some animal products all contain them.

Life is dependent on water. It facilitates the movement of nutrients throughout the body, the elimination of waste, and the control of body temperature.

23. Correct option (C)

When the body doesn't produce enough healthy red blood cells, it has anemia. Because red blood cells are responsible for delivering oxygen to the body's tissues, when there are not enough of them, the tissues do not receive adequate oxygen. Numerous symptoms, such as exhaustion, shortness of breath, pallor, and a racing heart, may result from this.

Anemia commonly manifests as fatigue and a lack of appetite. Pale skin is another typical symptom, but it can also be brought on by other illnesses.

Vitamin D deficiency is the root cause of rickets. Scurvy is a disease brought on by a deficiency of vitamin C. Vision loss is not a sign of anemia.

24. Correct option (D)

Vitamins that dissolve in fat are known as fat-soluble vitamins. They don't need to be refilled as frequently as water-soluble vitamins since they are kept in the body's fatty tissues. The fat-soluble vitamins include A, D, E, and K.

- Vitamin A is essential for vision, growth, and reproduction.
- Vitamin D is essential for bone health and immune function.
- Vitamin E is a powerful antioxidant that protects cells from damage.
- Vitamin K is essential for blood clotting.

25. Correct option (C)

Vitamins that dissolve in water are known as water-soluble vitamins. They must be replaced often via diet as the body cannot store them.

Vitamin C and the B vitamins are examples of water-soluble vitamins.

Collagen, a protein that provides skin, bones, and blood vessels their strength, is formed only when vitamin C is present.

The creation of energy, metabolism, and nerve function are just a few of the crucial body processes that the B vitamins play a role in.

On the other hand, vitamins that dissolve in fat are known as fat-soluble vitamins. The fat-soluble vitamins include A, D, E, and K.

Given that both vitamins B and C are water-soluble, the correct response is C.

26. Correct option (B & C)

Dietary fibers and water are both non-nutrients. They don't provide the body any calories or necessary nutrients. They can, nonetheless, nevertheless be healthy in other ways.

Water helps the body stay hydrated and is necessary for survival. Additionally, it facilitates the movement of nutrients and waste materials throughout the body. Dietary fibers support a healthy digestive tract and may lower the chance of developing certain chronic illnesses.

While lemons are a wonderful source of potassium, vitamin C, and other minerals. They also have dietary fibers in them. Lemons thus do contain nutrients.

27. Correct option (A)

A fat-soluble vitamin called vitamin D is necessary for a variety of body processes, including cell development, immune system health, and bone health. In the presence of sunshine, the body may produce vitamin D. The chemical transformation of a precursor molecule into vitamin D3 takes place when sunlight strikes the skin. Following absorption into the circulation and passage to the liver, vitamin D3 is transformed into calcidiol. The subsequent transformation of calcidiol into calcitriol, the active form of vitamin D, occurs.

28. Correct option (B)

Water aids in the digestion of meals, moves nutrients throughout the body, and maintains the digestive system's health, all of which aid in the body's ability to absorb nutrients.

Our body's ability to absorb nutrients is greatly aided by water. Many nutrients dissolve in water and are known as water-soluble nutrients. Water aids in the digestion of these nutrients and makes it easier for the small intestine to absorb them. Water also aids in the bloodstream's role in the distribution of nutrients throughout the body. Water is therefore necessary for healthy nutrition absorption and for maintaining general body processes.

Without adequate water, our bodies are less able to absorb nutrients, which can result in vitamin shortages.

29. Correct option (B)

a. Night blindness; iii. Poor vision in dim light.

Vitamin A is necessary for eyesight and its lack results in night blindness. Poor vision in low light is one of the signs of night blindness.

b. Beriberi; iv. Poor vision in dim light Patients.
Extreme weakness and a sour tongue are symptoms of beriberi.
Thiamine, a nutrient required for the creation of energy, is deficient in beriberi. Extreme weakness is one of the beriberi symptoms. Another sign of beriberi is a sour tongue, albeit it does not occur as frequently as acute weakness.
c. Goiter; I. Enlarged thyroid gland.
Iodine is required for the synthesis of thyroid hormones and its absence leads to goiter. An enlarged thyroid gland is one sign of goiter.
d. Anemia; ii. Deficiency of hemoglobin
Hemoglobin deficiency is the third kind of anemia. A lack of hemoglobin, a protein that transports oxygen in the blood, results in anemia. A lack of hemoglobin is one of the signs of anemia.
e. rickets; v. Brittle bones in children
Children with rickets have brittle bones. A lack of vitamin D, which is necessary for calcium absorption, results in rickets. Having brittle bones in youngsters is one of the signs of rickets.

LEVEL-II

1. Correct option (A)
A renowned research centre in India that specializes in the study of plant fossils and related areas is the Birbal Sahni Institute of Palaeobotany (BSIP). In the scientific field of paleobotany, plant fossils discovered in rocks, sediments, and other geological formations are identified, examined, and interpreted.
The institution was founded in 1946 and is named for Professor Birbal Sahni, a well-known Indian paleobotanist who made substantial contributions to the discipline. It is located in Lucknow, Uttar Pradesh, India.

2. Correct option (B)
Nutrients are a general term for the chemical elements found in food that give energy and are necessary for bodily growth and repair. The maintenance of several physiological systems and the promotion of general health depend on nutrients. They consist of water, vitamins, minerals, proteins, fats, carbs, and proteins, all of which have different functions within the body. These nutrients, which are required for optimal functioning and wellbeing, can only be acquired from the meals we consume.
Nutrients include things like proteins, carbs, and vitamins. They are not the only kind of nutrient, though.

3. Correct option (D)
Organic substances known as carbohydrates are made up of carbon, hydrogen, and oxygen atoms. Carbohydrates have the generic formula $(CH_2O)n$, where "n" denotes the molecule's number of carbon atoms. The ratio of carbon, hydrogen, and oxygen in carbohydrates is represented by this formula, which has the same number of hydrogen atoms as carbon atoms and half the amount of oxygen atoms. An essential part of carbohydrates, oxygen is necessary to both their structure and function.
Option (A), helium, is not an element found in carbohydrates. Helium is a noble gas that is not chemically reactive.
Option (B), lithium, is not an element found in carbohydrates. Lithium is a metal that is not chemically reactive.
Option (C), nitrogen, is found in some carbohydrates, but it is not the only element found in carbohydrates

4. Correct option (A)

Carbohydrates that you consume in excess of what your body requires will be stored as fat. Obesity and weight gain may result from this. A balanced diet that contains a range of foods, such as carbs, proteins, and fats, is crucial. You may do this to maintain a healthy weight and reduce your chance of developing obesity-related diseases.

Fats cannot be formed from proteins. The building components of biological tissues are called proteins. Using glucose as a fuel source, carbohydrates are converted to energy. Vitamins do not become fat-soluble. The body need vitamins as necessary elements for healthy health.

5. Correct option (A & B)

Both mangoes and rice are abundant in carbs. The body's primary fuel source is carbohydrate. They may be found in a wide variety of foods, such as cereals, fruits, vegetables, dairy products, and other foods.

Complex carbs may be found in rice, a grain. The body breaks down complex carbs gradually, releasing energy gradually. Fruits like the mango are a good source of simple carbs. Simple carbs provide the body a fast energy boost since they are easily broken down by the body.

Both rice and mango are nutritious foods that may be included in a diet that is balanced.

6. Correct option (C)

Amino acids, which are proteins' building components, make up proteins. Proteins are created by combining the 20 distinct amino acids. An amino group (-NH2) and a carboxyl group (-COOH) are chemical molecules that are linked to a central carbon atom to form amino acids. The structure and purpose of a protein are determined by the amino acid sequence.

Carbohydrates, or option (A), are not the constituent parts of proteins. Carbon, hydrogen, and oxygen make up the organic substances known as carbohydrates.

Fats in option (B) are not the fundamental components of proteins either. They are necessary for the growth and development of the body. Vitamins, option (D), are not the components that make up proteins.

7. Correct option (D)

Protein-rich diets are sometimes referred to as "bodybuilding" foods since they are essential for the body's repair and tissue growth processes. Our bodies convert the proteins in diets high in protein into amino acids, which are the building blocks needed for tissue development and repair. The phrase "bodybuilding" meals refers to how crucial this is for maintaining and growing muscles. In addition to producing enzymes and supporting the immune system and hormone synthesis, proteins also perform other vital bodily tasks.

Although option (A), "tissue-building," is a reasonable way to describe how proteins work, it is not the phrase used to specifically characterize foods that are high in protein.

Protein activity cannot be adequately described as energy-giving in Option (B). The body does not primarily obtain its energy from proteins.

The chemical constituents of proteins are known as amino acids, option (C). But amino acids aren't edible.

8. Correct option (A)

Protein is found in meat, milk, eggs, and pulses. For the development and repair of tissues, protein is a crucial nutrient. The synthesis of enzymes, hormones, and antibodies depends on it as well.

A good source of plant-based protein is pulses. They are an excellent source of iron, fiber, and folate as well. An excellent source of animal-based protein is milk. Additionally, it is a good source of potassium, calcium, and vitamin D. Eggs are a rich source of fat and protein

from animal sources. They are also an excellent source of selenium, choline, and vitamin B12. An excellent source of animal-based protein is meat. Additionally, it is an excellent source of zinc, vitamin B12, and iron.

9. Correct option (D)

Rich sources of fat include ghee, butter, and oil. Fatty acids, which are chains of carbon atoms connected by bonds, are the building blocks of all of them. When the body breaks down the fat, a lot of energy is released from these linkages.

A liquid fat called oil is obtained from either plants or animals. Made from milk, butter is a solid fat. By taking the milk particles out of butter, ghee is created, which is clarified butter. Due to the high calorie content of all three of these items, it is vital to eat them in moderation. But they may also be a beneficial component of a diet that is balanced.

10. Correct option (C)

There are 9 calories in 1 gram of fat. Compared to proteins and carbs, which both produce 4 calories per gram, this is more than twice as many calories.

Because fat is composed of fatty acids, which are chains of carbon atoms connected by bonds, it has a high calorie content. When the body breaks down the fat, a lot of energy is released from these linkages.

11. Correct option (D)

Heart disease is only one of the health issues caused by excess body fat. The inability of the heart to adequately pump blood is known as heart disease.

Extra body fat raises the risk of cardiovascular disorders like heart attacks, high blood pressure, and stroke and can cause heart issues. Elevated cholesterol levels, insulin resistance, and inflammation are a few risk factors linked to obesity that can put stress on the heart and circulatory system. Maintaining heart health requires maintaining a healthy body weight through a balanced diet and frequent exercise. Chest discomfort, breathing difficulties, and other issues may result from this.

Additionally, high blood pressure, high cholesterol, and diabetes can be brought on by excess body fat. Heart disease risk may also be increased by certain circumstances.

12. Correct option (B)

Based on the solubility of vitamins, vitamins are classified into two types: fat-soluble vitamins and water-soluble vitamins.

Water-Soluble Vitamins: Vitamins that are Water-Soluble: These vitamins may dissolve in water. They consist of the B-vitamins (thiamine, riboflavin, niacin, pantothenic acid, pyridoxine, biotin, folate, and cobalamin) and vitamin C. Water-soluble vitamins are quickly eliminated through urine and are not retained in significant quantities by the body. In order to maintain appropriate levels, frequent consumption is required.

Fat-Soluble Vitamins: Vitamins that are Soluble in Fat: These vitamins are soluble in fat. These vitamins are vitamin A, vitamin D, vitamin E, and vitamin K. Fat-soluble vitamins can be stored in the body's fatty tissues and liver and are absorbed together with dietary lipids. They can collect in the body and, if ingested in excess, may cause toxicity since they are not eliminated as quickly as water-soluble vitamins.

13. Correct option (C)

Calcium is a mineral that is essential for strong bones and teeth. It is also important for muscle function, blood clotting, and nerve transmission.

Calcium is a key element in the formation of bones. In order to guarantee appropriate bone development during infancy and adolescence, while bones are growing and developing, a sufficient intake of calcium is required.

Bone Strength: A sufficient calcium intake lowers the risk of osteoporosis, a disorder characterized by weaker and brittle bones, and helps prevent bone loss.

Teeth Health: Getting enough calcium helps build and maintain healthy tooth enamel, which lowers the risk of dental issues including cavities and sensitivity.
Phosphorus is also important for bones and teeth, but it is not as essential as calcium. Iron is important for red blood cell production, and iodine is important for thyroid function.

14. Correct option (A)

The indigestible component of plant meals is known as roughage or fiber. Dietary fiber, often known as roughage, is a component of plant-based meals such fruits, vegetables, whole grains, and legumes. It is made up of inedible plant cell wall components, namely cellulose, hemicellulose, and lignin.
It gives the faces greater volume and makes it easier for it to pass through the digestive system. This may aid in avoiding constipation.
Constipation cannot be avoided with vitamins, proteins, or lipids. Actually, certain fats might aggravate constipation.

15. Correct option (D)

The primary ingredient in sweat, urine, and faeces is water. It facilitates the removal of waste items from the body and controls body temperature
Sweat: The main component of sweat is water, with trace quantities of minerals and waste materials. Sweating aids in body temperature regulation and toxin removal.
Urine: The kidneys are crucial in the removal of waste and extra water from the circulation. Urine, the ensuing liquid waste, is made up of water and several dissolved materials, including urea, salts, and toxins.
Faeces: Water from the gut is absorbed when food passes through the digestive tract, resulting in the creation of solid waste. Constipation can be avoided and good bowel motions can be maintained with adequate drinking.
Sweat, urine, and faeces do not involve oil or ghee in their creation. Vitamins are necessary nutrients that support the body's ability to operate, but they are not involved in the production of these physiological fluids.
So, the correct option is D) Water

16. Correct option (A) ‘

Excessive water loss from the body results in dehydration. Sweating, nausea, diarrhoea, or a lack of fluid intake are all potential causes. Dizziness, weariness, headaches, and disorientation are just a few of the health issues that dehydration can cause. Dehydration may be lethal in extreme circumstances.
When water vapor in the sky condenses, it precipitates, falling to the earth as rain, snow, sleet, or hail. Water evaporation is the process through which water from the Earth's surface turns into vapor and enters the atmosphere.

17. Correct option (B)

A balanced diet is one that includes all the essential elements in the proper proportions. This indicates that the diet should consist of a variety of foods from all the food categories, including fruits, vegetables, grains, foods high in protein, and dairy.
For the purpose of sustaining excellent health, a balanced diet is crucial. It can aid in the prevention of chronic conditions including heart disease, stroke, type 2 diabetes, and several cancers. Additionally, it may aid in maintaining a healthy weight and enhancing general wellbeing.
A balanced diet is a healthy diet that includes all the necessary nutrients, while protein deficiency is a condition that occurs when the body does not get enough protein.

18. Correct option (B)

Malnutrition is defined as consuming either few or too many of the diet's important nutrients. When there is an imbalance, deficit, or excess of nutrients in the body owing to

insufficient dietary intake, poor nutrient absorption, or elevated nutritional needs, it is referred to as malnutrition.
Different symptoms of malnutrition may appear. Undernutrition is characterized by a lack of or insufficient intake of vital nutrients, which impairs growth and development and causes nutritional shortages. On the other side, overnutrition happens when a person consumes too many calories and other nutrients, which can cause obesity and other health issues.
Hence the correct option is B) Malnutrition

19. Correct option (C)

Kwashiorkor is a severe form of protein-energy malnutrition that develops when the diet is lacking in protein. Young children in impoverished nations that consume diets lacking in protein and other necessary nutrients are frequently affected.
Protein is essential for a number of physiological functions in the body, including growth, tissue repair, immunological function, and others. The body cannot achieve its protein requirements when there is insufficient protein consumption, which leads to the symptoms of kwashiorkor.
Edema (swelling), stunted development, muscular loss, exhaustion, irritability, changes in hair and skin color, and a compromised immune system are some of the signs of kwashiorkor. A well-balanced diet that offers enough protein and other necessary nutrients is used as treatment to alleviate the protein shortage.

20. Correct option (A & B)

A severe case of malnutrition known as marasmus is characterized by a lack of protein, carbs, and other vital nutrients. Children with chronic undernutrition and little access to a healthy food are most likely to experience it.
The development, maintenance, and repair of biological tissues depend on protein, whereas the energy needed for numerous internal processes comes from carbohydrates. Marasmus patients experience severe malnutrition overall, major weight loss, muscle wasting, compromised immune systems, stunted development, and other symptoms as a result of their diet's lack of protein and carbs.
It's critical to understand that marasmus is a complicated illness that involves deficits in more than simply protein and carbs. Additionally, marasmus sufferers lack other crucial components such vitamins and minerals.

21. Correct option (A)

A lack of vitamin B1, also known as thiamine, results in the illness known as beriberi. The neurological system and the metabolism of carbohydrates both depend on the vital vitamin thiamine. It is essential for turning food into energy.
Beriberi frequently develops in areas where polished rice is a staple meal but lacks the essential B vitamins. Wet beriberi and dry beriberi are the two main types. The cardiovascular system is impacted by wet beriberi, which increases the risk of edema and heart failure. Dry beriberi predominantly affects the neurological system and causes symptoms including muscular weakness, lack of coordination, tingling or numbness in the extremities, and trouble walking.
Vitamin B12, vitamin D, and vitamin K are not involved in the metabolism of carbohydrates. Therefore, the only option that is associated with beriberi is vitamin B1. So, the answer is A.

22. Correct option (B)

Scurvy is a condition that arises from a severe deficiency of vitamin C (ascorbic acid). Vitamin C is essential for the synthesis of collagen, a protein that helps in the formation of connective tissues, wound healing, and the health of blood vessels. When there is a lack of

vitamin C in the diet, collagen synthesis is impaired, leading to various symptoms associated with scurvy. The symptoms of scurvy include fatigue, weakness, bleeding gums, and skin problems.

Collagen formation does not include vitamin B1, vitamin D, or vitamin K. Vitamin C is the sole remedy therefore linked to scurvy. Therefore, B is the correct response.

23. Correct option (C)

The absence of adequate red blood cells in the body is known as anemia. Because red blood cells are in charge of transporting oxygen throughout the body, the organism suffers from an oxygen deficit when there are not enough of them.

Red blood cell formation requires vitamin B12, which is necessary. Anemia can develop if the body is unable to manufacture enough red blood cells due to a lack of vitamin B12.

Red blood cell formation does not include vitamin B1, vitamin C, or vitamin K. Therefore, vitamin B12 is the only remedy connected to anemia. The response is C.

24. Correct option (D)

A lack of vitamin D, which is necessary for healthy bone growth and development, can result in rickets. The body's ability to absorb and use calcium and phosphorus is greatly aided by vitamin D.

A vitamin D shortage results in poor mineralization of bones, which causes weaker and fragile bones. Children are frequently affected by rickets, which can result in skeletal abnormalities, stunted development, muscular weakness, and other associated problems.

Vitamins B1, C, and B12 do not contribute to the formation of bones. Therefore, vitamin D is the only supplement that has been linked to rickets. Therefore, D is the correct response.

25. Correct option (D)

A hallmark sign of goiter is enlargement of the thyroid gland. The term "goiter" describes the unnatural expansion of the thyroid gland, which is situated in the neck and generates hormones that control metabolism. Multiple reasons, including iodine shortage, certain drugs, autoimmune diseases, or underlying thyroid issues, might contribute to the growth.

The absence of adequate red blood cells in the body is known as anemia. Red blood cell synthesis is not influenced by the thyroid gland.

Vitamin insufficiency known as beriberi can harm the nerves. The manufacture of vitamin B1, which is lacking in beriberi, does not include the thyroid gland.

A deficiency of vitamin D can cause rickets, a disorder where the bones do not grow normally. The synthesis of vitamin D does not include the thyroid gland.

Therefore, the answer is D) Goiter.

3. Let us understand Plants

Solutions

LEVEL- I

1. Correct option (B)
Buds are small swellings on a plant from which a shoot, leaf, or flower usually develops. They are protected by a covering of modified leaves called scales which tightly enclose the more delicate parts of the bud.

2. Correct option (C)
Rootlets are also called lateral roots and are defined as the small branches of roots that form from the main root and helps the plant for the absorption of water and minerals as they increase surface area. So, the correct option is 'Lateral roots'.

3. Correct option (B)
Fibrous root system is an intricate network of similar sized roots with tiny hair like structures called root hairs. They form a mesh like bundle and does not penetrate deep into the soil. Examples: Maize, Wheat, Grass, etc.

4. Correct option (A)
The petiole is a stalk that connects the blade with the leaf base.

5. Correct option (D)
Stomata which are present on the lower surface of the leaves can also be called as nose of the leaves because through it plants take in Oxygen (O2) and Carbon Dioxide (CO2).

6. Correct option (B)
Ginger is an important and commonly used rhizome. It is a rhizome which is a modified stem of the plant. The rhizome of ginger bears nodes and internodes and scaly leaves are also present.

7. Correct option (C)
Stomata are the tiny, kidney, or bean-shaped pores or openings present in the epidermis of the cell.

8. Correct option (B)
Grasses are monocots and have a fibrous root system. They have a hollow, thin cylindrical stem and the leaves have parallel venation.

9. Correct option (B)
Essentially all the water used by land plants is absorbed from the soil by roots. A root system consists of a complex network of individual roots that vary in age along their length.

10. Correct option (B)
Beetroot forms a tuber that can be elongated, flat-round, round or egg-shaped depending on the variety.

11. Correct option (B)
Plants absorb water and minerals from the soil via their roots. The root hairs on roots increase the surface area of roots to facilitate absorption of water and minerals that are dissolved in the soil water.

12. Correct option (C)
Plants with fibrous roots have parallel venation. Plants that have fibrous roots and parallel venation in leaves include wheat, maize, grass, banana, bamboo, etc.

13. Correct option (B)
The flat green portion of a leaf is called as lamina or leaf blade.

14. Correct option (A)
Veins and veinlets are vascular bundles of lamina. The veins are larger and usually one prominent vein runs from base to apex of lamina and known as midrib. It forms lateral veins and the ultimate branches of veins are called as veinlets.

15. Correct option (B)
Reticulate type of venation is found in dicots hence the root system found in them will be the taproot system.

16. Correct option (C)
Transpiration is the process of removal of excess water from the plants through tiny openings in leaves known as stomata.

17. Correct option (D)
The main function of the leaf is photosynthesis. Photosynthesis is a physio-chemical process in which plants use light energy from the sun to drive the synthesis of organic compounds. Another important process carried out by leaf is transpiration. Transpiration refers to the evaporative loss of water by the stomata present in the leaves.

18. Correct option (D)
Underground stems are non-green stems that may take part in perennation, store food or help in vegetative propagation. Underground stems of potato, ginger, turmeric, Samarkand, Colocasia are modified stem to store food in them. Besides the loss of water vapour in transpiration, exchange of gases takes place through the pores called stomata present on the leaves.

19. Correct option (A)
Stem helps in transportation of food and water to different parts of the plant.

20. Correct option (B)
Midrib is the thickest vein present in the middle of the leaf. It separates the lamina or leaf blade into 2 parts. Smaller veins are observed radiating from the midrib.

21. Correct Matching is: a-Q, b-S, c-T, d-R, e-P

22. Correct Matching is: a-S, b-R, c-T, d-Q, e-P

LEVEL II

1. Correct option (D)
Products like timber, fodder, gums and resin, medicinal plants, fruits and vegetables, etc are procured from the plants and vegetation.

2. Correct option (B)
The plants which have soft and green stems are called herbs. Examples- Cauliflower, wheat, rice, mint etc.

3. Correct option (C)
Herbs are small plants with green and tender stems. The stem has very few branches. These plants have less strength in their stem and can easily be uprooted from the soil. Basil, coriander, Okra and thyme are a few examples of herbs around us.

4. Correct option (B)
Plants which have hard woody stems and branches close to the ground are called shrubs.

5. Correct option (C)
Rose is a shrub since it is a medium sized plant with a hard woody stem and branches out from the base.

6. Correct option (D)
The elongated stem of a tree that supports branches and has a woody outer bark is called the trunk of a tree. Trunk is only present in trees.

7. Correct option (A)
A plant having thin, long and weak stem which cannot stand upright and spreads on the ground is called a creeper.
8. Correct option (B)
Climbers are the plant that climbs on support by developing tendrils are modified stems that help to bind the support.
example-bougainvillaea, Money plant and Grapevine.
9. Correct option (C)
Roots help the plant to absorb water and nutrients from the soil and also provide support to the plant. The plants absorb water from the soil with the help of root hair through osmosis.
10. Correct option (A)
Taproot systems feature a single, thick primary root, called the taproot, with smaller secondary roots growing out from the sides.
11. Correct option (D)
A taproot system comprises one thick, cylindrical main root known as the primary root. Small roots that branch out from the primary root are termed as secondary and tertiary roots. Examples include Turnip, Parsnip, Beetroot, Carrot, Radish.
12. Correct option (A)
Carrots are the modified tap root, for storage. Carrots have a root cap, no nodes, and never directly bear leaves or flowers. Roots serve as food storage organs, absorbing nutrients and moisture, anchoring the plant in the soil, providing physical support for the stem, and anchoring the plant in the soil.
13. Correct option (B)
Aerial roots are roots above the ground. They are almost always adventitious. They are found in diverse plant species, including epiphytes such as orchids, tropical coastal swamp trees such as mangroves, banyan figs.
14. Correct option (D)
The shoot system refers to the parts of the plant growing above ground. This includes the stem, leaves, flowers, Twigs, and buds.
15. Correct option (C)
The stem may not always be typically like what they are expected to be. They are modified to perform different functions. Underground stems of potato, ginger, turmeric, Samarkand, Colocasia are modified to store food in them.
16. Correct option (B)
The fleshy stem of the cactus plant is used to store water. Cacti usually grow in very dry and hot conditions where there is a lack of water resources. Hence, they need to store sufficient water in their stem to meet their needs.
17. Correct option (A)
The spines or sharp needle-like structures seen in cacti are the modified leaves. The actual leaves are modified in order to reduce the surface area which will, in turn, reduce the evaporation of water through stomata.
18. Correct option (C)
In some plants with weak stems, the leaves are modified to form coiled thread-like structures called tendrils. The plants with weak stems can grow horizontally against a support as the tendrils coil around the support.
19. Correct option (C)
Leaves prepare food through the process of photosynthesis. So, they are called the food factories of plants. Photosynthesis is the process by which green plants and some other

organisms use sunlight to synthesize food in the form of glucose using carbon dioxide and water.

20. Correct option (C)

Lamina is the flat and green part of leaf. Lamina is also called leaf blade. Photosynthesis occurs in the lamina of leaf. Lamina contains many linear structures called veins which transport minerals and nutrients to leaf.

21. Correct option (A)

The arrangement of the pattern of veins in a leaf is called leaf venation.

22. Correct option (B)

When the veins in leaves are irregularly distributed to form a channel, it is known as reticulate Venation. Plants with reticulate venation in their leaves are likely to have tap roots.

23. Correct option (C)

Chloroplasts are chlorophyll containing plastids present in leaves of plants that functions as photosynthetic systems to produce sugars.

24. Correct option (A)

Photosynthesis is a process where plant leaves convert carbon dioxide and water into sugars and oxygen is released out in the environment.

25. Correct option (D)

Leaves of plants have small pores known as stomata which aids in gaseous exchange and loss of excess water through transpiration.

4. Nutrition in plants

Solutions

LEVEL – I

1. Correct option (A)

Roots are the plant organs that provide anchorage to the plants and are responsible for the uptake of water and nutrients from the soil & transporting it to the other plant parts. These are specialized and have tiny root hairs that help them to achieve their desired functions.

2. Correct option (B)

Stephen Hales was an English Botanist who first discovered the role of light in plant nutrition. Stephen Hales realized that air could be fixed by plants under the influence of solar light. Priestley showed that plants could regenerate the bad air making it breathable. Ingenhousz demonstrated that the green parts of plants performed the complete purification of air only under the influence of the light.

3. Correct option (A)

Starch is a carbohydrate which is synthesized by plant leaves and is stored as energy reserve by a process known as photosynthesis. To test the presence of starch, Iodine is frequently used. Iodine has the property to bind to Starch molecules and impart a blue-black colour which is an indication of presence of starch.

4. Correct option (A)

Potassium helps regulate the opening and closing of the stomata on leaves, which regulates the exchange of water vapor, oxygen, and carbon dioxide. Potassium also activates enzymes responsible for starch synthesis in plants.

5. Correct option (A)

In plants, minute pores are present on the surface of leaves that serves as the centre for gaseous exchange. The gases diffuse into the intracellular spaces of the leaf through these pores. Gaseous exchange (Air) is necessary for plants for the purpose of photosynthesis and respiration.

6. Correct option (B)

Parasitic plants and fungi have a root-like extension known as Haustorium, that grows into or around another plant to absorb water or nutrients from that plant instead of directly taking it from the soil. Parasitic plants like Dodder and Mistletoe have a haustorium.

7. Correct option (D)

Moist Bread and pickles if left outside, grows a black and white spot on the surface due to spoilage by Fungus, generally by Rhizopus and Mucor species. They generally grow on moist and damp habitat and in warm temperatures.

8. Correct option (A)

Transpiration is the loss of water in the form of water vapour through stomata, cuticles and lenticles. Most of the transpiration occurs through stomata in leaves i.e., 50-90% of total transpiration.

9. Correct option (A)

Urea serves as fertilizers and is a rich source of nitrogen. The urea reacts with the gases present in the soil and is firstly converted into ammonia and then into ammonium. This ammonium is then converted into nitrates and nitrites. Plant leaves turn yellow when there is a deficiency of nitrogen content. Farmers sprinkle urea so that they absorb

nitrogen and fulfill its needs. The leaves soon become green again, therefore increasing the rate of photosynthesis.

10. Correct option (D)

Drocera, Utricularia and Venus fly trap are insectivorous plants that feed on insects as well as their larvae. They generally have modified shoots especially for the purpose of capturing and ingesting the insect.

11. Correct option (B)

Cuscuta also known as Dodder or Amarbel is a genus of parasitic plants which has no chlorophyll, and it absorbs its food and nutrients from its host by wrapping its whole flexible stem around the host plant and absorbing nutrients and stays attached to it until the host dies.

12. Correct option (A)

Parasitic plants and fungi have root-like structures known as Haustorium, that grow into or around another plant and its tissues to absorb water and nutrients from that plant instead of directly taking it from the soil. Parasitic plants like Dodder and Mistletoe have a haustorium as specialised root like appendages projecting out from stem.

13. Correct option (D)

In the process of photosynthesis, Carbon dioxide, CO_2 along with Water, H_2O in presence of solar energy is converted to sugars, specifically Sucrose and O_2 inside the chloroplast present in the leaves of plants. Given below is the reaction.

$6CO_2+6H_2O \rightarrow C_6H_{12}O_6+6O_2$

14. Correct option (C)

Tropical Pitcher plants or Nepenthes is a genus of insectivorous plants that grows in Nitrogen deficient soil and feeds on insects. Nepenthes pitchers produce a lot of sweet nectar to attract insects.

15. Correct option (A)

Drocera is commonly known as Sundew, that are “flypaper” plants that trap prey in sticky hairs on their leaves. They make up one of the largest groups of carnivorous plants. Long tentacles protrude from their leaves, each with a sticky gland at the tip. These droplets look like dew glistening in the sun, thus their name.

16. Correct option (B)

Lenticels are specialized pores present on the surface of stems/ barks and permit the exchange of gases between the environment and the internal tissue spaces of the organs. They permit the entrance of oxygen and simultaneously the output of carbon dioxide and water vapor.

17. Correct option (A)

Stomata are the openings present on the surface leaves that function in exchange of gases in plants continuously. The gases diffuse into the intracellular spaces of the leaf through these pores. Gaseous exchange (Air) is necessary for plants for the purpose of photosynthesis and respiration.

18. Correct option (D)

Carnivorous plants such as Picher plants and Sundews grow in soil which is deficient in Nitrogen. Nitrogen is an essential component for plants to produce proteins and vitamins. Hence, they rely on insects for their nitrogen need.

19. Correct option (D)

Micronutrients are those nutrients that is needed in a small quantity by a body for plant metabolism. There are 7 essential plant nutrient elements defined as micronutrients

[boron (B), zinc (Zn), manganese (Mn), iron (Fe), copper (Cu), molybdenum (Mo), chlorine (Cl)]. They constitute in total less than 1% of the dry weight of most plants.

20. Correct option (C)

In the process of photosynthesis, Carbon dioxide, CO_2 along with Water, H_2O in presence of solar energy is converted to sugars, specifically Sucrose inside the chloroplast present in the leaves of plants and releases O_2 gas outside as by-product. Given below is the reaction.

$6CO_2 + 6H_2O \rightarrow C_6H_{12}O_6 + 6O_2$

21. Correct Matching is:a-S; b-P; c-T; d-R; e-Q

22. Correct Matching is:a-S; b-R; c-Q; d-P; e-T

LEVEL-II

1. Correct option (A)

The study of different components of diet including carbohydrates, fats, and protein of living organisms is known as science of nutrition.

2. Correct option (C)

Plant leaves have organelles called as chloroplast that synthesis their food requirements by a process known as photosynthesis, where light energy is converted to chemical energy.

3. Correct option (B)

Plants require essential nutrients for proper growth and development. There are in total 17 nutrients that are required by plants. Out of which 6 are macronutrients and others are micronutrients. These are Carbon (C), Hydrogen (H), Oxygen (O), Nitrogen (N), Phosphorus (P), Potassium (K), Sulfur (S), Calcium (Ca), Magnesium (Mg), Boron (B), Chlorine (Cl), Copper (Cu), Iron (Fe), Manganese (Mn), Molybdenum (Mo), Nickel (Ni), and Zinc (Zn).

4. Correct option (A & C)

In the process of photosynthesis, water and carbon dioxide are the precursor material for sugar synthesis by plants and in the process, oxygen is liberated. So, water as well as carbon dioxide cannot be the end product of it.

5. Correct option (D)

Chlorosis is the loss of the normal green coloration of leaves of plants due to loss of chlorophyll. It leads to the yellowing of leaves. This is caused by iron deficiency in lime-rich soils, or due to diseases or lack of light. Other elements that cause chlorosis are Nitrogen, Potassium, Zinc, Manganese, Sulphur etc.

6. Correct option (A)

Dry matter in leafy vegetables and proteins are rich in nitrogen and its molecules such as amines. Thus, Nitrogen is a very crucial mineral that helps in improving both the quantity and quality of that matter.

7. Correct option (D)

In monocots the Guard cell surrounding stomata are dumb-bell shaped, but in case of dicots they are kidney bean shaped.

8. Correct option (A)

Saprophytes are plants/fungi that feed on dead and decaying matter and have heterotrophic mode of nutrition. They lack chloroplast in their cells and thus they aren't green.

9. Correct option (C)

The leaves of the croton plant have chlorophyll, but they appear dark red in color. This is because the higher percentages of other pigments anthocyanins and carotenoid

present in them, hides the green color of the chlorophyll. So, the leaves of the croton plants can also make food by the process of photosynthesis.

10. Correct option (D)

Fungi are ubiquitous that is are found everywhere in this earth, majority of them can grow on the surface of the soil, foods items, in liquid, cloths and leathers, decaying matter etc.

11. Correct option (A)

Plants can synthesize their own food requirement by a process known as photosynthesis, and thus are autotrophic in nature.

12. Correct option (B)

Saprophytes are those organisms that feed and grow on dead and decaying matter. Fungi are such organisms that can grow and derive nutrients from dead leaves and humus in soil. Mushrooms are an example of fungus. Venus's flytrap and the Dodders are insectivorous, Lichen is a symbiotic association between algae and fungi.

13. Correct option (D)

Food that we consume has nutrients and minerals and serves as a mode of nourishment for the body organs for the purpose of energy to carry out metabolism.

14. Correct option (B)

Plant cells have chloroplast, in which stacks of thylakoids form grana and are suspended in matrix known as stroma. Thylakoids have chlorophyll inside which absorbs light and carries out photosynthesis.

15. Correct option (C)

Partial parasitic plants are green as they perform photosynthesis, but they are dependent on other plants for certain nutrients.

16. Correct option (B)

Crows draw their nutrition from both plant and animal sources. The organisms that draw their nutrition from both animals and plants are called omnivores. Hence, the crows are omnivorous organisms.

17. Correct option (A)

Starch is a polymer of sugars present in plants, its presence is detected by iodine, iodine binds to starch turns it into blue-black color.

18. Correct option (B)

Plant hormones are not nutrients but chemicals that in small amounts promote and influence the growth, development, and differentiation of cells and tissues.

19. Correct option (D)

Plant leaves appear green due to chlorophyll present in their chloroplast. Chloroplasts are the sites of photosynthesis in plants and chlorophyll serves as a light absorbing pigment.

20. Correct option (A)

During Photosynthesis, plants use solar energy to convert water and carbon dioxide into sugars and oxygen is released and this sugar is then used as chemical energy during respiration.

21. Correct option (D)

The end product of photosynthesis is Glucose, $C_6H_{12}O_6$ which is formed from water and carbon dioxide in the presence of light.

22. Correct option (B)

The associations of organism where they live together and share shelter and nutrients and are mutually beneficial for each other is known as Symbiosis. Lichen is an example of Symbiosis.

23. Correct option (A)
Except for Mango rest all are heterotrophs, that is they cannot synthesize their own food and must derive their nutrition from other sources.

24. Correct option (D)
Iron and magnesium are essential for the synthesis of chlorophyll in plants. A deficiency of these elements causes chlorotic patches on leaves with the leaves turning yellow.

25. Correct option (B)
They can trap insects because they do not grow in nitrogen-rich soils. Soil lacking nitrogen causes pitcher plants to get nitrogen from the trapped insects.

26. Correct option (A)
The soil of marshy land is deficient in nitrogen. Plants living in marshy areas do not get nitrogen from the soil. To fulfil their nitrogenous need, they need to eat insects.

27. Correct option (B)
These tiny pores present in the leaves are known as stomata. They help in the exchange of gases and give out water vapor through transpiration.

28. Correct option (D)
The associations between organisms where they live together and share shelter and nutrients and are mutually beneficial for each other is known as Symbiosis. Lichen is an example of Symbiosis.

29. Correct option (A)
Rhizobium is a nitrogen fixing bacteria present in the soil that converts atmospheric nitrogen into ammonia which can be taken up by plant roots.

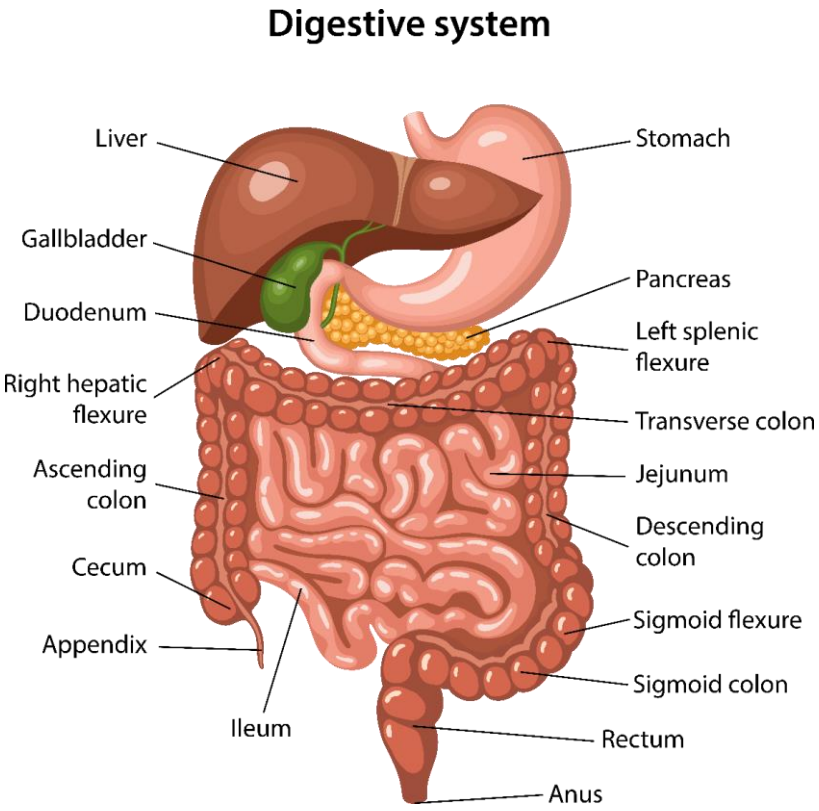
30. Correct option (A)
All green plants have chlorophyll in their cells to prepare their own food by photosynthesis. During the process of photosynthesis, water combines with carbon dioxide to form sugars and oxygen in liberated in the process.

5. NUTRITION IN ANIMALS

Solutions

LEVEL – I

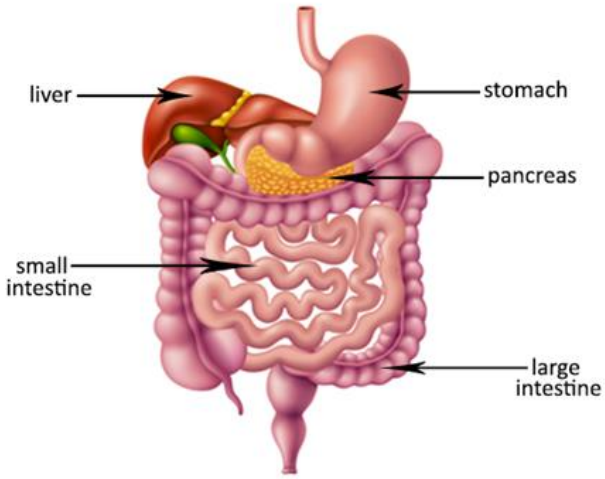
1. Human beings ingest food through the mouth. Digestion begins in the mouth when we chew food. The salivary glands present in the mouth secrete their digestive juice called the saliva. Saliva breaks down the starch present in the food and converts it into sugar. The saliva also keeps the food moist making it easy to swallow. The food that we ingest goes down from the mouth through the oesophagus into the stomach. Stomach receives the partly digested food coming through the food pipe. Digestion of food in the stomach takes one to four hours, depending on the type of food eaten. Digestion of food happens with the help of various secretions such as hydrochloric acid and gastric juice secreted by the walls of the stomach. The semi-digested food, called chyme, leaves the stomach and enters the small intestine. When the food enters the small intestine, it mixes with the bile and the pancreatic juice, secreted into it by the liver and the pancreas, respectively. The small intestine also produces digestive juices, which help to complete the digestion of all the components of food. Here absorption of food takes place in the small intestine. The food that remains undigested and unabsorbed then enters the large intestine. The function of the large intestine is to absorb water and some salts from the undigested and unabsorbed food. The semi-solid food that is left behind is then stored as faeces in the rectum, the last part of the large intestine. The semi-solid faeces is removed from our body from anus.



2. Ruminants have a special stomach with four chambers, namely, rumen, reticulum, omasum and abomasum. The swallowed food goes straight into the rumen—the first and the largest of the four chambers in the stomach. The food is partially digested in the rumen. The partially digested food in the rumen is known as cud. The cud is pushed into the reticulum—the second chamber of the stomach and from here it is sent back into the mouth for thorough chewing. The chewed food is swallowed and sent into the omasum—

the third and the smallest chamber of the stomach to absorb excess water from the food. In the omasum, the food is broken down to even smaller pieces. The last chamber—abomasum is called the true stomach. Gastric juices are secreted in the abomasum to help in digestion. Enzymes act upon the food to complete the digestion. The digested food then goes into the small intestine, where absorption of nutrients takes place. A large sac-like structure called caecum present between the small intestine and large intestine also contain bacteria that help in cellulose digestion.

3. 1-e, 2-a, 3-d, 4-b, 5-c
4. False
5. True
6. True
7. True
8. False
9. Ingestion, digestion, absorption, assimilation, and egestion are the five stages of holozoic nutrition.
10. Assimilation
11. The salivary gland secretes digestive juice called saliva which breakdown starch into sugars. It also keeps the food moist and easy to swallow.
12. The antelope swallows grass as the first chamber of its stomach called rumen contains cellulose digesting bacteria which digests the cellulose present in the grass. This is sent to the second chamber and then regurgitated back to the mouth where it is thoroughly chewed.
13. Bile juice
14. I am amoeba. I form a pellet of the undigested food and throw it out of the food vacuole through a temporary opening of the cell membrane.
15. Enzyme
16. Digestion
17. Mouth/Buccal cavity
18. I am any cooked grain such as rice or wheat. I test positive the first time as there is starch present. The second time, I test negative as the saliva breaks down my starch into sugar when I am chewed.
19. The image is villi present on the inner walls of a part of the small intestine. There are millions of finger-like projections of villi which absorb food. They increase the surface area for absorption of food.
20. Pancreas and liver are the two digestive glands. Their secretions are found in the small intestine.



21. Correct option (B)
The whole process starting from ingestion of food by mouth, its breakdown, absorption, and utilization by the body is known as digestion. It's a process by which complex food is taken up by the body.

22. Correct option (B)
Amoeba ingests its food with the help of its false feet or pseudopodia. The food is digested in the food vacuole.

23. Correct option (B)
Human saliva contains salivary amylase which breakdown starch into maltose and dextrin.

24. Correct option (B)
The hydrochloric acid in the gastric juice breaks down the food and the digestive enzymes split up the proteins. The acidic gastric juice also kills bacteria. The mucus covers the stomach wall with a protective coating.

25. Correct option (B)
Digestion and absorption of all biomolecules are completed in the intestine. The small intestine has 3 parts namely- the duodenum (c- shaped), jejunum, and ileum. Complete digestion of food takes place in the ileum. After being digested the food is absorbed by the blood vessels present in the small intestine.

26. Correct option (C)
The millions of finger-like outgrowths on the inner surface of the small intestine are called as Villi. They help in increasing the surface area for the absorption of the digested food.

27. Correct option (C)
Although the small intestine is narrower than the large intestine, it is the longest section of your digestive tube, measuring about 7.5m on average, or three-and-a-half times the length of your body.

28. Correct option (B)
Your body uses amino acids to make proteins. The different types of amino acids and the way they're put together determine the function of each protein.

29. Correct option (C)
Fats are present in the form of large globules in the small intestine. The small intestine receives the secretions from the liver and the pancreas. The bile salts (from the liver) break down the large fat globules into smaller globules so that the pancreatic enzyme lipase can easily act on them.

30. Correct option (D)
Water from the undigested food is absorbed in the large intestine to the body.

31. Correct option (D)
Protein after digestion by proteinases is converted into Amino acids which are absorbed into the bloodstream.

32. Correct option (D)
Digestion of groundnut oil produces fatty acids and glycerol. Groundnut oil is rich in fatty acids.

33. Correct option (D)
The boiled rice and saliva are added to the iodine solution then the colour of iodine will not change because of the digestion of starch present in rice into monomers of sugars.

34. Correct option (B)
Canines are used for ripping and tearing food such as meat.

35. Correct option (D)
Incisors – These are the four teeth which are present in both the upper and lower jaw known as the incisors. These are having the one root. The major function of the incisors is to cut food because they have the sharp edge.

36. Correct option (C)
4 incisors, 2 canines, 4 premolars and 6 molars (Each half of the Jaw has 2,1,2,3 number of teeth's, 2 Incisors, 1 Canine, 2 premolars and 3 molars on each side of the Jaw)

37. Correct option (B)
The root contains blood vessels and nerves, which supply blood and feeling to the whole tooth. This area is known as the "pulp" of the tooth.

38. Correct option (D)
Excessive intake of foods containing sugar causes tooth decay. The germs in the mouth form plaque on the surface of teeth

39. Correct option (C)
Amoebae use pseudopodia, which are arm-like projections on the cell surface, to engulf and ingest food.

40. Correct option (D)
Amoeba is a protist that has pseudopodia or false feet which it uses for locomotion and capturing of food. It has no fixed shape, no fixed mouth and possess no digestive system.

41. Correct option (C)
Peristalsis is a series of wave-like muscle contractions that move food through the digestive tract. It starts in the esophagus where strong wave-like motions of the smooth muscle move balls of swallowed food to the stomach.

42. Correct option (B)
Amoeba is a unicellular organism. It undergoes intracellular digestion that is food is engulfed inside, fused with lysosome and broken down by enzymes inside the cells. The mode of nutrition in Amoeba is holozoic. The process of obtaining food by Amoeba is called phagocytosis.

43. Correct option (B)
Premolars and Molars are the teeth that do most of the chewing because those flat surfaces are perfect for grinding and crushing food until it's ready to be swallowed.

44. Correct option (A)
Holozoic Nutrition is a process in which complex food is taken into a specialist digestive system and broken down into small pieces to be absorbed. This consists of 5 stages, ingestion, digestion, absorption, assimilation, and egestion.

45. Correct Answer: Liver
Bile is a fluid that is made and released by the liver and stored in the gallbladder.

46. Correct Answer: Small intestine
Stomach is connected to Duodenum of small intestine at its lower portion.

LEVEL - II

1. Correct option (C)
Hookworms are parasites that live inside the intestines. There, they feed on blood from the intestinal wall, mature and lay eggs.

2. Correct option (C)
The bees climb onto or into the flower and suck up the nectar with their straw-like mouth and collect it in a little sac called a crop.

3. Correct option (D)
Snakes do not have the right kind of teeth to chew their food so they must eat their catch whole. Their jaw is structured in such a way that it allows the mouth to open wider than their own body to swallow their prey whole.

4. Correct option (A)
Saliva is an extracellular fluid produced and secreted by salivary glands present in the mouth.

5. Correct option (D)
The food vacuole containing the food molecule fuses with lysosome which contains amylase and proteinase enzymes. These enzymes digests carbohydrates and proteins present in the food.

6. Correct option (B)
Hydrochloric acid is the main component of gastric juice and is secreted by the parietal cells or oxyntic cells of the stomach walls in humans.

7. Correct option (D)
The vacuole is pushed deeper into the amoeba by cytoplasmic movements where it fuses with lysosomes - organelles containing digestive enzymes like amylase, cellulase and proteinases which can digest carbohydrates such as Sugars and cellulose and proteins.

8. Correct option (C)
The finger-like outgrowths (Pseudopodia) of Amoeba help ingest food. However, the finger-like outgrowths or villi of human intestines increases the surface area of the walls and thus help to absorb the digested food.

9. Correct option (B)
Cud is the name given to the food of ruminants which is swallowed and partially digested. Cud is a food component that returns from the stomach to the mouth of an animal for a second chew. Rumination is the process by which cud is generated during digestion. Ruminants are other names for cud-chewing animals.

10. Correct option (A)
The large intestine has 3 primary functions: absorbing water and electrolytes, producing and absorbing vitamins, and forming and propelling feces toward the rectum for elimination.

11. Correct option (C)
Digestion is the mechanical and chemical break down of food into small organic fragments.

12. Correct option (D)
In a holozoic mode of nutrition, the organism feeds on complex organic matter. They ingest the complex organic matter, digest, and absorb the nutrients. In animals' mode of nutrition is holozoic because they depend on plants for food, cannot synthesize their own food, thus are heterotrophs, and they swallow their food.

13. Correct option (D)
Lacteals are lymph vessels responsible for absorbing fat in the small intestine. Occurring in the finger-like villi of the ileum, lacteals have a milky appearance and drain into the lymphatic system. Before fat can pass into the lacteal, bile from the liver causes its emulsification into small droplets for action of the enzyme lipase.

14. Correct option (B)
Deciduous teeth are the official term for baby teeth, milk teeth, or primary teeth. Deciduous teeth start developing during the embryonic stage and then commonly begin to come in about 6 months after birth.

15. Correct option (C)
The false feet of Amoeba or the pseudopodia are used to engulf and capture food and also in movement and locomotion.

16. Correct option (B)
The first set of teeth which are lost between the ages of **6 -12 years**, are termed as the milk teeth.

17. Correct option (D)
There are four types of teeth Incisors, Canines, Premolars and Molars.
18. Correct option (A)
In stomach, the medium is acidic with a pH of about 1-2 due to presence of Hydrochloric acid in the Gastric juice. It serves as suitable medium for digestion of proteins in the stomach.
19. Correct option (D)
The removal of faeces through anus is called excretion.
20. Correct option (B)
Premolars and molars pair of teeth differ in structure but are similar in function that is grinding of food.
21. Correct option (A)
Salivary Amylases digest starch into smaller molecules, ultimately yielding maltose, which in turn is cleaved into two glucose molecules by maltase.
22. Correct option (C)
Chyme is a churned semi-solid food that is formed in the stomach of the digestive tract.
23. Correct option (B)
The stomach is a thick-walled muscular bag. Its shape is like a flattened U and it is the widest part of the alimentary canal.
24. Correct option (D)
Although the small intestine is narrower than the large intestine, it is actually the longest section of your digestive tube, measuring about 7.5m on average, or three-and-a-half times the length of your body.
25. Correct option (C)
Proteins are chains of amino acids linked together, once they are broken by proteinases, they yield small units of amino acids.
26. Correct option (D)
Plenty of boiled and cooled water with a pinch of salt and sugar dissolved in it. This is called Oral Rehydration Solution (ORS).
27. Correct option (B)
Euglena has both an Autotropic as well as heterotrophic mode of nutrition. It's a protozoon that can synthesize its own food in presence of sunlight and can capture and consume its prey when deprived of sunlight.
28. Correct option (D)
The tip of the tongue is most sensitive to sweet and salty substances, the side to sour, while the back is most sensitive to bitter taste.
29. Correct option (A)
Villi are finger-like projections in the small intestines that increases surface area for absorption of digested food. They have very thin walls and have network of thin and small blood vessels close to the surface.
30. Correct option (B)
The swallowed food moves downwards in the alimentary canal because of the contraction of muscles in the wall of the food pipe. This movement is under the control of nervous system and is known as peristalsis.